# DH$^2$A-KT: Leakage-Audited Knowledge Attribution for Knowledge Tracing

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**Abstract**

Graph-augmented knowledge tracing (KT) often constructs auxiliary knowledge without a leakage audit, then reports a single AUC as if every graph component contributed. This paper asks a leakage-audited attribution question: which structural and temporal knowledge survives a controlled ablation? DH$^2$A-KT is an attribution pipeline that is tractable on a single consumer GPU (the $^2$ marks heterogeneous hypergraph *interfaces*; the recommended tracer does not use hypergraph convolution). It reuses the train-only protocol of companion paper P0, extends the audit to hyperedges, and then ablates which knowledge predicts. Pre-specified gates ($\Delta$val $\geq +0.002$ vs. a matched twin) show that Linear $\log(1+\Delta t)$ carries next-step accuracy on this backbone—including on a capacity-matched zero-incidence twin (5/5 PASS; mean $\Delta$val+0.00255)—whereas zeroing only the train-only question–KC incidence *message* fails (0/5 PASS; mean $\Delta$val$\approx -0.0001$), as do hyper-

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graph convolution, hint-item hyperedges, a full Q-matrix, teacher grouping, an expert DAG, and per-concept exponential forgetting. An older hard-off twin that removed the whole question pathway is capacity-mismatched and is not used to credit incidence. On XES3G5M fold 0 under a clean protocol (mask-repeats, chunked windows, $L$=400, test-only), the recommended tracer (QKC-T: LSTM with the Q$\leftarrow$KC pathway and a linear $\log(1+\Delta t)$ branch) reaches test AUC 0.8296±0.0001 over five training seeds versus compute-matched GIKT 0.8258±0.0001 and *simpleKT* 0.8214±0.0007 (five-seed mean gaps +0.0038 and +0.0082; paired learner-level bootstrap 95% CIs exclude 0). The same recipe on folds 1–2 stays within 0.001 (mean 0.8295±0.0004). On this backbone the exponential forgetting form (P4) does not outperform Linear $\log(1+\Delta t)$; history-only LSTM $\Delta t$ fails the gate (2/5 PASS). A graph-only diagnostic encoder (AUC 0.752) is in the appendix, not the recommended tracer.
*Keywords:* Knowledge tracing, knowledge attribution, data leakage, educational data mining

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## 1. Introduction

Knowledge tracing (KT) estimates a learner's latent mastery of knowledge components (KCs) from interaction history, supporting adaptive practice, intelligent tutoring, and curriculum recommendation. The field has progressed from probabilistic and sequence models [1] to graph-augmented models that propagate information over auxiliary concept graphs [2, 3], and, most recently, toward hypergraph representations that capture higher-order relations among knowledge components [4, 5, 6] and dynamic, temporallyevolving graph structure [7].

This paper builds on [8]'s train-only leakage protocol. DH$^2$A-KT is a leakage-audited *attribution* setting: construct, audit, then ask which knowledge predicts. The recommended tracer (QKC-T) is a GIKT-family LSTM with Q$\leftarrow$KC incidence and a linear $\Delta t$ branch (Section 4.3); hypergraph convolution fails that gate (Section 5.5). Teacher, forum, and video members remain interfaces. Fig. 1 is the pipeline.

### *1.1. Motivation*

Three concrete gaps motivate this work:

- **Higher-order structure is under-audited in graph-KT.** Hypergraph-KT models [4, 5, 6] group multiple knowledge components into a single hyperedge, yet published systems rarely audit whether those hyperedges were constructed without leakage, and few expose an interface for entities beyond KCs (hints, teachers, forums, video) when such logs become available.

- **Graph provenance is rarely audited.** [8] shows that pooling a knowledge-component graph from the full interaction log before a learner-based train/test split lets held-out information reach a graph-augmented KT model even when the sequence optimizer itself trains only on the training fold. [8]'s protocol audits this channel for *pairwise* prerequisite/similarity edges; published hyperedge-level audits remain scarce.

### *1.2. Research Questions*

This work is motivated by a broader question: *which leakage-audited knowledge actually improves next-step KT?* We decompose this into two questions, plus entities this paper does not claim to have learned:

- **RQ1** (§5.5): Under a leakage audit, which structural knowledge improves next-step AUC?
- **RQ2** (§6.2): On this backbone, does a Linear $\log(1+\Delta t)$ branch outperform the per-concept exponential forgetting form we test, and is the surviving timing gain available as LSTM-side attempt context?

An exploratory Critic-gated survey over frozen scores is reported after the predictive results (Section 6.4); it is not a third research question and not a contribution. Teacher, forum/discussion, and video nodes are defined in the schema (Section 3) but remain data-blocked on every corpus we use (Section 7.3): this paper establishes an interface for those entities, not empirical evidence of how they improve KT. A graph-only encoder and three-fold manipulation check remain available as a forced-reliance diagnostic (Appendix Appendix A); they are not RQ1's recommended system.

### *1.3. Contributions*

- **C1 (construction and audit).** A procedure that groups [8]'s audited, acyclic prerequisite edges into *path-derived concept-group* chain hyperedges (unordered member sets along directed paths; Section 4.1) and names three leakage classes that a pairwise projection does not express (Section 4.2); session+Hint hyperedges on FoundationalASSIST [9] (Section 6.7). Discussion-thread and teacher-intervention types remain data-blocked (Section 7.3). Construction is not claimed as an AUC lift.

- **C2 (which knowledge predicts).** A pre-specified ablation (gate $\Delta$val $\geq +0.002$ vs. a matched twin; an internal convention locked before the ladder, not a public registry) plus a clean-protocol tracer: Linear $\log(1+\Delta t)$ on the QKC-T backbone passes (5/5 PASS; test AUC 0.8296±0.0001 on XES3G5M fold 0, mean 0.8295±0.0004 across three learner folds), and the same timing branch also passes on a capacity-matched zero-incidence twin (5/5 PASS; Table 6). A capacity-matched zero-incidence twin for the incidence *message* fails (0/5 PASS; Table 5), so incidence content is not credited. HypergraphConv, hint-item hyperedges, a full Q-matrix, teacher grouping, an expert DAG, and per-concept exponential forgetting (P4) also fail (Table 9). Versus compute-matched GIKT 0.8258±0.0001, the five-seed mean gap is +0.0038 and a paired learner-level bootstrap 95% CI excludes 0 (Table 4). Graph-only v2 mean AUC 0.752 versus GKT 0.834 is an appendix diagnostic, not the recommended tracer.

## 2. Related Work

### *2.1. Graph- and hypergraph-based knowledge tracing*

Pairwise KT graphs [2, 3] and dynamic variants [7] propagate over edges; hypergraph KT [4, 5, 6] and heterogeneous node types [10, 11] still draw members from KC/exercise/learner sets. Among the hypergraph-KT systems we read [4, 5, 6, 10, 11], Teacher, Hint, Forum, and Video appear as interfaces here rather than as evaluated members. This paper's quantitative core is a GIKT-class LSTM with audited Q$\leftarrow$KC incidence (Section 4.3); hypergraph operators are pre-specified failures (Table 9).

### 2.2. Forgetting and temporal effects in KT

Sequence KT models have long treated elapsed time as a forgetting cue. [12] extend DKT with multiple forgetting-related features (lag from the previous interaction and past trial counts). HawkesKT [13] models skill-to-skill temporal cross-effects with a Hawkes-process kernel (WSDM 2021). Our P4 arm is narrower: a per-concept exponential $\exp(-\mathrm{softplus}(\theta_c) \cdot \mathrm{gap})$ on the same v4 backbone as Linear $\log(1+\Delta t)$. A FAIL of that one form does not rule out power-law decay, data-driven half-lives, or Hawkes kernels; we do not claim to have tested those alternatives.

### 2.3. LLM agents and explanation faithfulness

Closest in scope is [14], which aligns LLM mastery reports with implicit DLKT states and evaluates explanation faithfulness with KC-overlap, an independent-generation control, and paired ablations. Section 6.4 is a scoped survey in that line, not a third pillar: we gate rationales with a Critic over *frozen* $P$(correct) and use ID-anchored KC-Jaccard because XES3G5M has no in-repo synonym KC vocabulary. Broader LLM-agent surveys [15, 16, 17, 18] motivate the frozen-core split: re-inferring mastery with a multi-agent LLM over millions of interactions is not tractable on one consumer GPU.

### 2.4. Leakage-controlled evaluation

We searched ACM Digital Library, IEEE Xplore, and Scopus through July 2026 with the terms "hypergraph knowledge tracing", "leakage audit", and "graph provenance". [8] (companion P0, under review) audits pairwise prerequisite/similarity edges under train-only construction. Within that search we found no published protocol that applies the same leakage diagnostics to *hyperedges*; this paper extends the audit from pairwise edges to hyperedges (Section 4.2). We reuse P0's train-only construction, four pairwise diagnostics, and pinned artefacts (Section 3.1).

Observational treatment-effect estimators in education [19, 20, 21] are background only.

## 3. Preliminaries

Let $\mathcal{D} = \{(u, t, q, c, y)\}$ be a KT interaction log with learner $u$, timestamp $t$, item $q$, knowledge component $c$, and correctness label $y \in \{0, 1\}$, following [8]'s notation. A learner-based temporal split $\mathcal{F} = (\mathcal{F}_{\text{train}}, \mathcal{F}_{\text{val}}, \mathcal{F}_{\text{test}})$ assigns each learner to exactly one fold (ratios 0.7/0.1/0.2, three folds with split seeds 42/43/44, matching [8]'s protocol so results remain directly comparable, see Section 5.1).

### *3.1. Self-contained summary of the reused P0 protocol*

Because [8] is under review, we state the reused pieces here. For each learner-based fold, $E^f_{\text{pre}}$ and $E^f_{\text{sim}}$ are built from *train-fold interactions only*, then cycle-pruned to a DAG. Four pairwise diagnostics (P0 Table of leakage metrics; computed on the train-only graph vs. held-out learners) are:

- $\text{ECR}^{\text{flag}}$: structural learner-overlap indicator (whether any learner appears in more than one split);
- $\text{ECR}^{\text{overlap}}$: held-out pattern overlap on retained edges;
- $|\rho|$: magnitude of the Pearson correlation between edge weight and held-out outcome;

- TBMR (P0 also writes TBVR): *within-train* temporal mixing — the share of post-cutoff *training* mass that supports retained prerequisite transitions — *not* a train/test membership violation.

P0 reports that switching from full-log to train-only graph construction shifts baseline AUC by at most 0.003 on XES3G5M, ASSISTments 2012, and Junyi. We pin the companion repository to commit `5e3d6a0` and copy its published baseline CSVs verbatim (Section 5.2). The companion PDF can be attached as journal supplementary material on request. Chain hyperedges in this paper are bounded-length paths on the same audited $E^f_{\text{pre}}$ (Section 4.1).

We extend this notation with a hyperedge $h = (\kappa, \{(\tau_i, e_i)\}, f)$: kind $\kappa \in$ {path-derived concept-group, session, discussion-thread, teacher-intervention}, typed members (student, exercise, concept, teacher, hint, forum, discussion, video), and fold $f$. Chain hyperedges on audited $E_{\text{pre}}$ are audited and used in FAIL/diagnostic arms; they do *not* enter the recommended tracer (Table 4). Teacher/forum/video kinds are interfaces; session write-backs appear only in the Tier 2 pilot. The recommended tracer consumes train-only Q←KC incidence and timestamps inside the recommended tracer (Section 4.3).

## 4. Tier 1: DH$^2$-KT

### *4.1. Heterogeneous hypergraph construction*

Path-derived concept-group hyperedges are bounded-length simple directed paths on audited, acyclic $E^f_{\text{pre}}$ (Algorithm 1). Each hyperedge collects the concepts visited along a directed path, but Algorithm 1 emits an *unordered* member set, so path direction is not retained in the hyperedge rep-

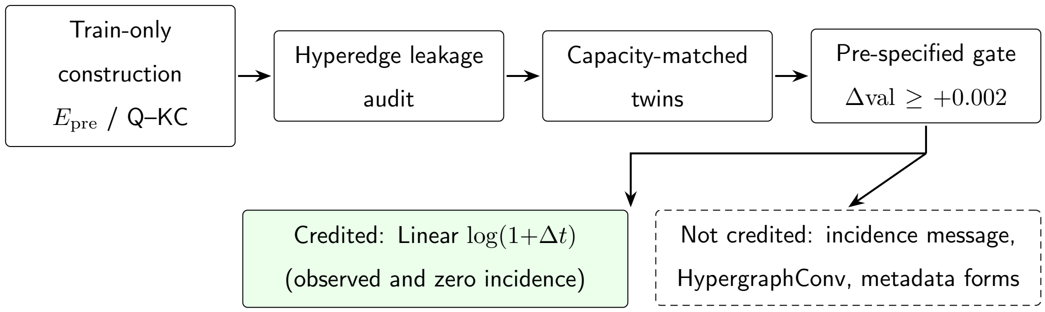


Figure 1: DH$^2$A-KT is an attribution pipeline, not a single predictor. Train-only knowledge is constructed and audited, then compared against capacity-matched twins. Only Linear $\log(1+\Delta t)$ survives the credit gate on XES3G5M; the question–KC incidence message and the tested hypergraph/metadata forms do not. QKC-T is the observed-incidence evaluation arm; Zero+$\Delta t$ is the minimalist positive control. The dashed graph-only encoder and the exploratory Critic survey are not part of this credited map (Appendix A, 6.4).

resentation. This grouping is used for audit and for optional `HypergraphConv` diagnostic arms; it is not part of the recommended Q←KC tracer.

**Input:** $E^f_{\mathrm{pre}}$; bounds $[\ell_{\min}, \ell_{\max}] = [3, 8]$
**Output:** path-derived concept-group set $\mathcal{E}^{\mathrm{path}}_f$
Build directed graph $G$ from $E^f_{\mathrm{pre}}$
**for** each root $r$ with in-degree 0 in $G$ **do**
  **for** each simple path $p$ from $r$ with $\ell_{\min} \leq |p| \leq \ell_{\max}$ **do**
    emit unordered members $\{(\mathrm{concept}, c) : c \in p\}$
  **end for**
**end for**

**Algorithm 1.** Path-derived concept-group hyperedges from audited $E^f_{\mathrm{pre}}$ (member set is unordered; path direction is not encoded). Chain length bounds $[3, 8]$ match `hyperedge.concept_prerequisite` in `configs/xes3g5m.yaml`.

Discussion-thread and teacher-intervention hyperedges are data-blocked on the public KT corpora we use. Session+Hint hyperedges are built on FoundationalASSIST only (Section 6.7). Chain hyperedges on XES3G5M are audited and used in the graph-only diagnostic (Appendix Appendix A), not as the recommended tracer's message-passing graph.

### *4.2. Hyperedge-level leakage audit*

The four pairwise diagnostics of Section 3.1 are reused on the pairwise projection of a hyperedge set. Pairwise TBMR/$|\rho|$ are skipped when that projection exceeds the implementation cap (Table 2). Table 1 adds three classes that a pairwise projection does not express.

Measured values on XES3G5M fold 0 appear in Table 2. To verify that the audit can register leakage when it is present, we add a **full-log positive control**: chain hyperedges built from $E_{\text{pre}}$ pooled before the learner split, with a concept→interaction map from the entire log. Under this deliberately contaminated construction, group-membership leak rate reaches 1.0 while the train-only arm stays at 0; pairwise TBMR on a $10^5$-pair subsample is similar across arms ($\approx 0.73$), confirming that the zero train-only flags reflect protocol design rather than a blind audit.

### *4.3. Recommended tracer: LSTM with Q←KC and time*

The recommended tracer, denoted QKC-T, is an LSTM backbone with train-only question–KC incidence and a linear $\log(1+\Delta t)$ branch; no convolution is applied on the concept stack. (Implementation flags `-question-graph`

Table 1: Hyperedge-level leakage taxonomy (extends [8] Table 1)

| Class | Source | Diagnostic |
| --- | --- | --- |
| Group-membership | A hyperedge with $\geq 1$ member drawn from a held-out interaction while other members are train-only | Group-membership leak rate (XES3G5M fold 0: 0; Table 2) |
| Cross-modal | A Forum/Video embedding pooled over content timestamped after the prediction cutoff | Not instantiated (no Forum/Video logs on the corpora we train) |
| Intervention-timing | A Teacher/Hint hyperedge whose content was effectively chosen using knowledge of the student's later correctness | Not instantiated for Teacher; Hint members on FoundationalASSIST use train-only session construction (group-membership leak 0) |

and `-no-graph` turn incidence on and concept-stack `HypergraphConv` off; Table 3.) Path-derived concept-group chain hyperedges are not consumed on this path. Let $A \in \mathbb{R}^{N_q \times N_c}$ be the train-only question–KC incidence, row-normalized to $\bar{A}$. Each item $q$ has an embedding $e_q$; each KC $c$ has $e_c$. One propagation step, in the spirit of GIKT [3] rather than hypergraph convolution, is

$$\tilde{e}_q = \tanh\big(W_q e_q + W_c \, \bar{A}_{q,:} \, E_c\big), \tag{1}$$

Table 2: Path-derived concept-group hyperedge audit on XES3G5M fold 0. **Train-only** uses P0's audited $E^{f}_{\text{pre}}$ and a train-only concept→interaction map. **Full-log** is a deliberate positive control: $E_{\text{pre}}$ pooled before the learner split with a full-log member map (GS Hau A4). Pairwise TBMR/$|\rho|$ use a random sample of $10^5$ projected pairs (seed 42) when the full projection exceeds $10^5$ pairs.

| Metric | Train-only | Full-log (pos. control) |
|---|---|---|
| Hyperedges | 446,305 | 588,369 |
| $\text{ECR}^{\text{flag}}$ | 0.0 | 0.0 |
| Group-membership leak rate | 0.0 | **1.0** |
| TBMR (sampled) | 0.734 | 0.736 |
| $\|\rho\|$ (sampled) | 0.0 | 0.0 |

implemented with a single shared matrix ($W_q{=}W_c{=}W$) on the sum $e_q + \bar{A}_{q,:}E_c$, then a separate input projection into the LSTM. We do not claim a non-collapsible dual-map form: the tied-$W$ implementation is equivalent to one linear on the sum. Attribution of the incidence *message* uses a capacity-matched twin that keeps the same question pathway and zeros only $\bar{A}$ (`-question-incidence-control zero`); an older hard-off twin that removed `question_embed` and the projection pathway is capacity-mismatched and is reported only as a pathway-off baseline (Table 4). Interaction embeddings of $(c_t, y_t)$ plus Rasch-style item parameters feed an LSTM. The next step is scored by reading a query vector at the *known* identity of item/KC $t{+}1$ (standard next-step KT: the model is told *what* will be asked, not whether the learner will be correct).

When `-time-gap` is on, a dedicated linear map of $\log(1 + \Delta t)$ can be added on the LSTM input, on the next-step query, or both. We treat the prediction time as *attempt-onset correctness prediction*: the score is produced when the learner opens item $t{+}1$, whose identity is already the query, not when the system first recommends that item. LSTM-side $\Delta t$ uses only gaps between already-observed steps (*history-only*). The query-side gap is the wait until the scored item; that item's identity is part of the prediction request, so the gap is attempt context rather than a post-hoc peek at duration after $y_{t+1}$ is observed. We do not feed the scored step's own response into the query. We report a history-only $\Delta t$ row (`-time-gap-mode lstm`): five-seed mean $\Delta$val+0.00193 (2/5 PASS; Table 4, Table 10). Query-only fails on all five seeds (mean +0.00099); only `both` passes 5/5. Per-concept exponential forgetting is a pre-specified alternative that fails the gate (P4). HypergraphConv on multi-KC item hyperedges, hint-item hypergraphs, dual-gated graph-only $x_t$, and the graph-sensitivity auxiliary loss $L_{\text{aux}}$ belong to the v2 diagnostic encoder (Appendix Appendix A) or to FAIL arms (Table 9), not to Eq. (1).

## 5. Experimental setup

Table 3 maps development labels that appear in logs to the architecture names used in the text and table captions.

### *5.1. Datasets*

Following [8], XES3G5M [22] is the primary predictive benchmark: 18,066 learners, 7,652 items, 865 KCs and 5,549,635 recorded attempts, which expand to 6,413,353 KC-level rows under the standard multi-KC expansion (Section 5.3). ASSISTments 2012 GKT AUC from P0 is not reused as a comparison column. FoundationalASSIST [9] is a secondary fold with synchronized hints (Section 6.7), not interchangeable with classic ASSISTments 2012. Junyi supplies an expert DAG for structural overlap only (Section 6.6); we do not train the full tracer there.

### 5.2. Baselines reused from P0

Sequence baselines (DKT [1], AKT [23], simpleKT [24], GIKT [3], DyGKT [7], DGEKT [25]) are trained with the vendored pyKT implementations [26] under our clean chunked protocol (Section 5.3); we treat them as fair comparisons, not as novel KT architectures discovered here. We reuse [8]'s published results for BKT [27], GKT [2], and SKT [28] under train-only graph construction (vendored CSVs at commit `5e3d6a0`), except GIKT, AKT, simpleKT, DKT, DyGKT, and DGEKT on XES3G5M, which we retrain under the clean $L$=400 protocol (Table 4). GIKT shares DH$^2$-KT's compute envelope (batch 16, epoch cap 30, patience 5); AKT and simpleKT inherit pyKT/P0 defaults with only $L$ raised to 400. That is a matched *compute* budget, not a matched tuning search (Appendix Appendix B; cf. [29]). A config-checksum script confirms the XES3G5M P0 config match. The companion PDF is available to reviewers as supplementary material on request.

### 5.3. Clean evaluation protocol

Headline AUCs use a *clean* protocol aligned with pyKT's repeat-format warning [26, 24]: the library's KC-level export expands each multi-KC question into consecutive rows that share the same timestamp and answer, so a large share of eval positions are answerable by copying the input label. We therefore mask repeat-row targets (`-mask-repeats`) in both our models and retrained pyKT baselines, use chunked windows (`-window-mode chunked`) with $L$=400, and report test-only AUC on the remaining scored positions. Table 4 uses a shared pyKT-aligned intersection of $n_{\text{scored}}$=1,093,720 positions across all listed models. Native DH$^2$ scoring of the same clean mask yields 1,093,755 positions (difference of 35); calibration (Table 11) reports the native count. We do not collapse multi-KC rows back to attempt level for the headline AUC: each masked KC-row target is scored once. The 6,413,353 KC-expanded rows versus 5,549,635 recorded attempts in [22] is the same multi-KC expansion pyKT warns about; we score the masked remainder, not every expanded row. Published P0 numbers that score every KC row are not comparable to this table. Our contribution is a fair retrain under the shared masking rules, not the discovery of KC-level row repeats.

### *5.4. Credit gate*

A knowledge type is credited only if the pre-specified $\Delta$val $\geq +0.002$ vs. its matched twin. This is an internal convention, not a timestamped public registry. Five-seed validation AUC SD is 0.00015 on the recommended tracer and 0.00023 on Q←KC (Table 4 arms); +0.002 is ≈9–13× those SDs. For Q←KC incidence the matched twin is capacity-matched zero incidence (Table 5): mean $\Delta$val=$-0.00010\pm0.00034$ over five seeds (0/5 PASS), so incidence content is not credited. A historical hard-off comparison that removed the whole question pathway is capacity-mismatched and is not a credit test. Timing is credited twice: Linear $\Delta t$ vs. Q←KC mean +0.00274 ($t_4$=23.99, two-sided $p$=1.79×10$^{-5}$), and Linear $\Delta t$ on the zero-incidence backbone (Table 6) mean +0.00255±0.00015 (5/5 PASS). Holm–Bonferroni at $\alpha$=0.05 among the remaining XES credit arms leaves the QKC-backbone timing test significant. The six FAIL rows in Table 9 remain single-seed and are judged only by the +0.002 gate, not by $p$-values: the gate is a credit threshold anchored to seed SD, not a claim of simultaneous significance.

### *5.5. Manipulation check*

A hypergraph ablation is interpretable only if destroying the constructed hyperedges moves AUC. Near-total node-drop $p$=0.90 on the graph-only diagnostic encoder (Appendix Appendix A) confirms *dependence* on some graph input. Degree-preserving rewiring asks whether that encoder also uses *which* concepts co-occur: it passes on folds 0–2 ($\Delta$AUC 0.0318/0.0341/0.0351; clean AUC $\approx$ 0.75, matched to node-drop; Table A.19). Relation-label permutation is near-null on every fold (expected for a single-kind encoder). An earlier pairwise+exercise path was graph-inert ($\Delta$AUC$\approx$ 0.0003). The recommended tracer does not take this check: concept-stack `HypergraphConv` is off, and Q$\leftarrow$KC incidence is a bipartite membership matrix rather than a new multi-way graph.

Table 3: Architecture names used in this paper. Flags in parentheses are implementation aliases from training logs, not model names.

| Name in this paper | Architecture |
|---|---|
| $DH^2A$-KT | Attribution pipeline: train-only construction, leakage audit, matched twins, credit gate, and knowledge map; not one predictor |
| QKC-T reference arm | LSTM with train-only Q←KC incidence and Linear $\log(1+\Delta t)$; concept-stack HypergraphConv off (`v4`, `-question-graph`, `-no-graph`) |
| Q←KC only | Same LSTM without the $\Delta t$ branch |
| Pathway-off twin | Same LSTM with the question pathway removed (historical hard-off; capacity-mismatched) |
| Capacity-matched zero twin | Same question pathway with incidence matrix zeroed (`-question-incidence-control zero`) |
| Zero-incidence+$\Delta t$ (minimal positive control) | Capacity-matched zero twin plus Linear $\log(1+\Delta t)$ at LSTM and query (`zero_time`) |
| Graph-only diagnostic | HypergraphConv encoder on chain hyperedges, no question LSTM path (`v2`) |

## 6. Results

### *6.1. Which structural knowledge predicts (RQ1)*

Table 4 reports *test-only* AUC on XES3G5M fold 0 under the clean protocol (mask-repeats, chunked windows, $L$=400). Table 4 lists fold 0 point estimates; Table 8 adds learner-fold and training-seed stability. Over five training seeds (A2 matrix), DH$^2$-KT with the Q←KC pathway reaches 0.8270±0.0002 versus compute-matched GIKT 0.8258±0.0001, AKT 0.8223±0.0001, and *simpleKT* 0.8214±0.0007 (all retrained under the same masking rules). Retrained DKT and DGEKT are 0.8009±0.0002 and 0.7850±0.0001; the pathway-off baseline is 0.8206±0.0002. Under the capacity-matched zero-incidence twin (Table 5), the incidence *message* fails the gate on all five seeds (0/5 PASS; mean $\Delta$val=−0.00010±0.00034). Learner-level bootstrap (1000 resamples, 3614 learners) for Q←KC minus GIKT is +0.00117, 95% CI [+0.00078, +0.00153] (excludes 0). Adding Linear $\log(1+\Delta t)$ reaches test AUC 0.8296±0.0001 (5/5 PASS vs. the no-$\Delta t$ twin; five-seed mean gap vs. GIKT +0.0038; a paired learner-level bootstrap 95% CI [+0.00342, +0.00413] excludes 0). We do not credit hypergraph convolution for these numbers: that arm is `-no-graph` on the concept stack. The same Q←KC+$\Delta t$ recipe on folds 1 and 2 stays within 0.001 of fold 0 (mean 0.8295±0.0004 over learner folds; 0.8296±0.0001 over five training seeds on fold 0; Table 8).

Table 9 lists pre-specified failures ($\Delta$val gate +0.002, not loosened). HypergraphConv on multi-KC item hyperedges (K), hint-item session hyperedges on FoundationalASSIST (L), a full Q-matrix (M1), teacher grouping (M4), a Junyi expert DAG on a 5k-user subsample (M5), and per-concept exponential forgetting (P4) all FAIL. The small FoundationalASSIST session-

HE shift 0.720 → 0.722 (concept projection; Table 15) is therefore not a hypergraph-AUC claim: the isolated hint-item HE arm did not pass.

### *6.2. Time as attempt context (RQ2)*

On XES3G5M, Linear $\Delta t$ into both LSTM and next-step query passes against Q←KC on all five training seeds (5/5 PASS at the pre-specified $\Delta$val $\geq$ +0.002; test-only 0.8296±0.0001 vs. 0.8270±0.0002, mean difference +0.00261, learner-level 95% CI $[+0.00225, +0.00298]$). The same timing branch also passes when the backbone has *zero* Q–KC incidence (5/5 PASS; mean $\Delta$val+0.00255, mean $\Delta$test-only+0.00269; Table 6), so the timing credit is not an artefact of nonzero incidence. A quartile slice of log(1+gap) collapses to two bins: short gaps ($\lesssim$18 min) $\Delta$+0.0028, long gaps $\Delta$+0.0019, both CIs excluding 0 — not a long-horizon exponential-decay pattern under P4's form. Table 10 ablates injection site on XES3G5M. History-only LSTM $\Delta t$ (observed gaps, no query-side wait) reaches five-seed test AUC 0.8288±0.0002 and fails as a credit claim (mean $\Delta$val+0.00193, 2/5 PASS); query-only fails (0/5, mean +0.00099); only `-time-gap-mode both` passes on all five seeds (mean +0.00274). The branches are complementary: the recommended tracer's timing credit is not available under a history-only deployment. On FoundationalASSIST, query-only $\Delta t$ keeps most of the M2 gain (+0.01471 vs. off) while LSTM-only is +0.00280; combining time with prior-step `saw_answer` is additive (+0.00297 vs. M2). Per-concept $\exp(-\mathrm{softplus}(\theta_c) \cdot \mathrm{gap})$ fails versus Linear (P4). We treat Linear $\Delta t$ as attempt-context knowledge on this backbone, not as a hypergraph effect, and we do not generalise that to every forgetting form. Query-side $\Delta t$ uses the gap to the item whose identity is already the next-step query; it is not the duration of a step whose correctness has been observed. Against AKT at the same $L$=400 clean protocol (test AUC 0.8223±0.0001 over five seeds), the five-seed mean gaps are +0.0074 versus AKT and +0.0082 versus *simpleKT*. A paired learner-level bootstrap (1000 resamples, 3614 learners) on one canonical fold-0 prediction file (not the subtraction of those ± rows) gives 95% CIs $[+0.00703, +0.00792]$ vs. AKT and $[+0.00916, +0.01013]$ vs. *simpleKT*; vs. GIKT the paired CI is $[+0.00342, +0.00413]$ (Table 4).

### *6.3. Calibration (A6)*

Table 11 reports 15-bin ECE, Brier score, and NLL on fold 0 test predictions under the same clean protocol. The recommended tracer is best calibrated (ECE 0.0056) despite higher AUC; Q←KC on is slightly miscalibrated high (ECE 0.0103). We report calibration as a diagnostic, not as a primary claim.

### *6.4. Exploratory survey: Critic-gated explanations*

This subsection is a scoped survey after the Tier 1 results, not a contribution. Three agents read frozen DH$^2$-KT scores: a Diagnostician writes a rationale, a Critic flags output that contradicts the frozen $P$(correct) (15-bin ECE 0.0056 on the recommended tracer; Table 11), and a Tutor writes session hyperedges that the frozen tracer does not consume. We ran $n$=500 prompts (XES3G5M fold 0, seed 42, Qwen2.5-7B via Ollama) on the graph-only v2 checkpoint and again on the recommended tracer. Following [14], we report Critic flag rate and ID-anchored KC-Jaccard. Independent generation (IG) omits $P$(correct) from the Diagnostician prompt while the Critic still sees the true score, so the flag-rate gap is largely definitional and is not a content

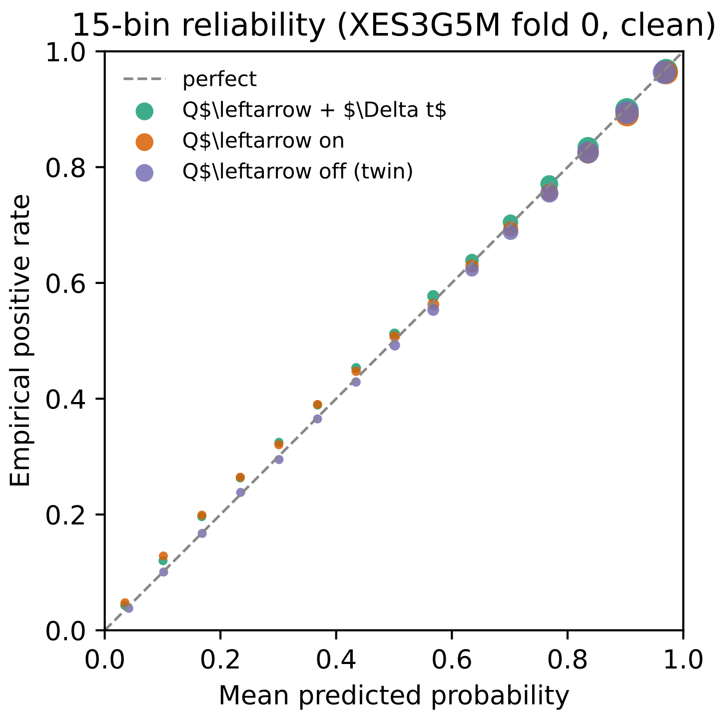


Figure 2: Fifteen-bin reliability on XES3G5M fold 0 under the clean protocol (mask-repeats, chunked, $L$=400; seed 42 checkpoints). Marker area scales with bin count. The QKC-T checkpoint stays closest to the diagonal (ECE 0.0056; Table 11). Three DH$^2$-KT arms are shown; pyKT baselines were not re-scored for this diagram.

finding. On the recommended tracer, grounded/IG flags are 0.018/0.766 with mean JC 0.161/0.152; the JC gap is $\approx +0.010$ (Table 12, Table 13). Blinded Likert ratings on a seed 42 subsample of $n$=100 grounded tracer explanations (3 independent raters) give faithfulness 4.55±0.63 and usefulness 4.01±0.74 (ordinal Krippendorff $\alpha$=0.52/0.56; mean pairwise Spearman $\rho$=0.43/0.34; within±1 95.0%/94.7%; Table 14). An earlier dual-rater $n$=40 pack on the v2 graph-only checkpoint is archived and not the primary human-eval row.

### 6.5. Paired diagnostic knobs

Table 15 pairs remaining diagnostic knobs with the survey's faithfulness knobs. Removing the Critic sets the flag rate to 0 by construction while leaving ID-anchored KC-Jaccard unchanged. Dropping $L_{\text{aux}}$ (`graph_sensitivity_weight` = 0) on the graph-only fold 0 track still yields clean AUC 0.754 and $\Delta$AUC= 0.039 (pass); that track is the appendix diagnostic, not Table 4.

### 6.6. Ground-truth structural overlap (Junyi)

Table 16 compares hyperedge-chain overlap with P0's pairwise $E_{\text{pre}}$ against the Junyi expert DAG at $K$=|expert|. Edge F1 is 0.165 (hyperedge chain) vs. 0.183 ($E_{\text{pre}}$) on fold 0; folds 1–2 are analogous. The chain representation slightly *underlaps* pairwise $E_{\text{pre}}$: collapsing directed paths into multi-node hyperedges, then projecting back to pairs for overlap scoring, can dilute expert-edge recall relative to scoring $E_{\text{pre}}$ directly. We treat this as structural validation of construction fidelity, *not* as evidence that chain hyperedges improve predictive KT on Junyi — full DH$^2$-KT training on Junyi (25M interactions) was not run.

### *6.7. Secondary corpus: FoundationalASSIST*

FoundationalASSIST [9] provides ≈1.74M skill-linked interactions with synchronized `hint_count` (hint-use rate 5.2%). Using the same learner-temporal protocol as P0 (0.7/0.1/0.2, fold 0), we infer train-only $E_{\text{pre}}$ (280 edges after cycle pruning), build 5,840 path-derived concept-group chain hyperedges, and construct 308,670 session hyperedges on train users (11.6% with a Hint member; $\text{ECR}^{\text{flag}}=0$, group-membership leak=0). DH$^2$-KT fold 0 AUC is 0.720 with path-derived concept-group hyperedges alone ($n$=298,500 predictions). Adding up to 20k multi-concept session hyperedges (Hint-preferring selection; concept co-occurrence projection) raises AUC to 0.722 ($\Delta$+0.002) — a small positive shift under the same local budget, not a matched GKT comparison. Table 17 summarizes the corpus and those AUCs. An IPW identification check of hint use, with weak positivity, is in Appendix Appendix C; it is not a contribution.

Table 4: Test-only AUC on XES3G5M fold 0 under the *clean* protocol: mask-repeats, chunked windows, max seq. len. 400. All models are scored on an identical set of 1,093,720 target positions. Every row is mean±SD over five training seeds (42, 17, 1234, 0, 2024). DH$^2$-KT uses a v4 LSTM with question embeddings; Q←KC is bipartite incidence (`-question-graph`), not hypergraph convolution; $+\Delta t$ is Linear log(1+gap) into LSTM and next-step query (`-time-gap`, both). The pathway-off twin removes the question pathway (capacity-mismatched); the capacity-matched zero-incidence check is Table 5 and fails the gate. The history-only row is `-time-gap-mode lstm` over five seeds (Table 10): mean $\Delta$val fails the gate (2/5 PASS). pyKT [26] baselines are retrained under the same export and masking rules; P0 published numbers use a different (leaky) protocol and are not comparable. GIKT is matched to the DH$^2$ *compute* envelope (batch 16, cap 30 epochs, patience 5), not to a hyperparameter search (Appendix Appendix B). Five-seed mean gaps on this table are +0.0038 vs. GIKT, +0.0074 vs. AKT, and +0.0082 vs. *simpleKT*. Paired learner-level bootstrap CIs on one canonical fold-0 file (not those mean gaps) exclude 0 (Section 6.2). DyGKT at $L$=400 failed to converge (valid AUC ≈0.60); a clean rerun at $L$=200 reached 0.6756 and is omitted here. This is *not* the three-fold graph-only v2 table (Appendix Appendix A).

| Model | Test AUC | $n$ scored |
|---|---|---|
| DH$^2$-KT Q←KC $+\Delta t$ | 0.8296±0.0001 | 1,093,720 |
| DH$^2$-KT Q←KC $+\Delta t$ history-only | 0.8288±0.0002 | 1,093,720 |
| DH$^2$-KT Q←KC | 0.8270±0.0002 | 1,093,720 |
| GIKT (compute-matched) | 0.8258±0.0001 | 1,093,720 |
| AKT ($L$=400) | 0.8223±0.0001 | 1,093,720 |
| DH$^2$-KT pathway-off twin | 0.8206±0.0002 | 1,093,720 |
| *simpleKT* ($L$=400) | 0.8214±0.0007 | 1,093,720 |
| DKT ($L$=400) | 0.8009±0.0002 | 1,093,720 |
| DGEKT ($L$=400) | 0.7850±0.0001 | 1,093,720 |

Table 5: Capacity-matched Q←KC attribution on XES3G5M fold 0 (clean protocol). Both arms keep `-question-graph` (same `question_embed`, GCN linear, and input projection); only the train-only incidence matrix is replaced by zeros in the twin (`-question-incidence-control zero`). Pre-specified gate: $\Delta$val $\geq +0.002$. Checkpoint state shapes match on every seed; observed nonzero links = 8848, zero arm = 0. Mean $\Delta$val=$-0.00010 \pm 0.00034$; mean $\Delta$test-only$\approx +0.00004$. 0/5 PASS: the incidence *message* is not credited. The older hard-off twin that removed the whole question pathway is not used for this claim (Section 5.4). $\Delta$test columns are outer test-only AUC (1,093,755 scored positions).

| Seed | Observed val | Zero val | $\Delta$val | $\Delta$test-only | Gate |
|---:|---:|---:|---:|---:|---|
| 0 | 0.82554 | 0.82610 | −0.00056 | −0.00007 | FAIL |
| 17 | 0.82579 | 0.82589 | −0.00010 | −0.00018 | FAIL |
| 42 | 0.82598 | 0.82619 | −0.00021 | +0.00010 | FAIL |
| 1234 | 0.82605 | 0.82567 | +0.00038 | +0.00014 | FAIL |
| 2024 | 0.82608 | 0.82610 | −0.00003 | +0.00020 | FAIL |

Table 6: Linear $\Delta t$ on the capacity-matched *zero-incidence* backbone (XES3G5M fold 0, clean protocol). Both arms keep zero Q–KC incidence; `zero_time` adds LSTM+query $\log(1+\Delta t)$ (`-time-gap-mode both`). Pre-specified gate: $\Delta$val $\geq +0.002$. Mean $\Delta$val=$+0.00255 \pm 0.00015$; mean $\Delta$test-only=$+0.00269 \pm 0.00012$. 5/5 PASS: timing credit is not an artefact of nonzero incidence. State shapes differ by the timing parameters (expected).

| Seed | Zero val | Zero+$\Delta t$ val | $\Delta$val | $\Delta$test | Gate |
|---:|---:|---:|---:|---:|---|
| 0 | 0.82610 | 0.82852 | +0.00242 | +0.00270 | PASS |
| 17 | 0.82589 | 0.82835 | +0.00245 | +0.00258 | PASS |
| 42 | 0.82619 | 0.82862 | +0.00243 | +0.00255 | PASS |
| 1234 | 0.82567 | 0.82837 | +0.00270 | +0.00282 | PASS |
| 2024 | 0.82610 | 0.82882 | +0.00272 | +0.00280 | PASS |

Table 7: Leakage-audited knowledge attribution summary. Multi-seed XES3G5M rows use five paired seeds; other rows are pre-specified single-seed diagnostics and are not significance tests. The pre-specified credit convention is $\Delta$val $\geq +0.002$ versus the stated twin. "Observed" and "zero" refer only to the Q–KC incidence message; both retain the same question embedding and projection pathway.

| Knowledge message | Matched twin | Corpus / seeds | Mean $\Delta$val | Credit |
|---|---|---|---|---|
| Q←KC incidence | Zero incidence | XES / 5 | −0.00010±0.00034 | 0/5 FAIL |
| Linear $\Delta t$ (both) | No $\Delta t$, observed incidence | XES / 5 | +0.00274±0.00025 | 5/5 PASS |
| Linear $\Delta t$ (history-only) | No $\Delta t$, observed incidence | XES / 5 | +0.00193 | 2/5 FAIL |
| Linear $\Delta t$ (both) | No $\Delta t$, zero incidence | XES / 5 | +0.00255±0.00015 | 5/5 PASS |
| HypergraphConv | No concept convolution | XES / 1 | +0.00025 | FAIL |
| Hint-item session HE | No hint-item HE | FA / 1 | +0.00007 | FAIL |
| Full Q-matrix incidence | Matched control | FA / 1 | −0.00010 | FAIL |
| Teacher grouping | Matched control | ASSIST2012 / 1 | +0.00142 | FAIL |
| Expert prerequisite DAG | Matched control | Junyi / 1 | +0.00056 | FAIL |
| Per-concept exponential gap | Linear $\Delta t$ | XES / 1 | −0.00258 | FAIL |

Table 8: Q←KC+$\Delta t$ test-only AUC on XES3G5M under the clean protocol (mask-repeats, chunked, $L$=400). **Rows 0–2:** one model per learner fold (P0's three split seeds); mean±SD is over folds. **Bottom row:** training-seed stability on fold 0 ($n$=5 seeds: 42, 17, 1234, 0, 2024; A2 matrix); mean±SD is over seeds.

| Fold | Test AUC | $n$ scored |
|---|---|---|
| 0 | 0.8296 | 1,093,720 |
| 1 | 0.8298 | 1,092,664 |
| 2 | 0.8291 | 1,090,374 |
| **Mean±SD (folds)** | **0.8295±0.0004** | — |
| Fold 0, 5 training seeds | **0.8296±0.0001** | 1,093,720 |

Table 9: Negative results under a pre-specified validation gate ($\Delta$val $\geq +0.002$ vs. a matched twin). All runs keep a separate Linear/embedding branch (no absorption into `concept_embed`). FAIL means the knowledge type is *not* credited as predictive. Hypergraph convolution and hint-item hyperedges are among the failures; they are not AUC novelty.

| ID | Knowledge (corpus) | Twin val | ON val | $\Delta$val | Gate |
|---|---|---|---|---|---|
| K | HypergraphConv, multi-KC HE (XES) | 0.82598 | 0.82623 | +0.00025 | FAIL |
| L | Hint-item session HE (FA) | 0.80861 | 0.80868 | +0.00007 | FAIL |
| M1 | Full Q-matrix incidence (FA) | 0.80834 | 0.80824 | −0.00010 | FAIL |
| M4 | Teacher grouping (ASSIST2012) | 0.76435 | 0.76577 | +0.00142 | FAIL |
| M5 | Expert prereq. DAG (Junyi 5k) | 0.7640 | 0.76456 | +0.00056 | FAIL |
| P4 | Per-concept exp. forgetting (XES) | 0.82878 | 0.82620 | −0.00258 | FAIL |

Table 10: Query-side vs. history-only Linear $\Delta t$ on XES3G5M fold 0 (clean $L$=400, five training seeds). Twin is Q←KC without time-gap. Pre-specified gate: $\Delta$val $\geq +0.002$. `lstm` is the history-only arm (observed gaps only). Mean±SD test AUC over seeds; gate column is $k/5$ PASS. Only `both` passes on all five seeds.

| Mode | Injection site | Test AUC | Gate ($k/5$) |
|---|---|---|---|
| `both` | LSTM + query | 0.8296±0.0001 | 5/5 (mean $\Delta$val +0.00274) |
| `lstm` (history-only) | LSTM only (observed gaps) | 0.8288±0.0002 | 2/5 (mean $\Delta$val +0.00193) |
| `query` | next-step query only | 0.8279±0.0002 | 0/5 (mean $\Delta$val +0.00099) |

Table 11: Calibration on XES3G5M fold 0 test predictions under the clean protocol (mask-repeats, chunked, $L$=400). ECE uses 15 equal-width bins; lower is better. Checkpoints: seed 42 A2 matrix (twin AUC 0.8202 here; Table 4 reports the five-seed mean 0.8206±0.0002). Native DH$^2$ scoring yields $n$=1,093,755 positions; the headline AUC table uses the pyKT-aligned intersection $n$=1,093,720 (difference of 35). Reliability curves appear in Figure 2.

| Model | AUC | Brier | NLL | ECE |
|---|---|---|---|---|
| Q←KC +$\Delta t$ | 0.8296 | 0.1207 | 0.3828 | 0.0056 |
| Q←KC on | 0.8270 | 0.1216 | 0.3857 | 0.0103 |
| Q←KC off (twin) | 0.8202 | 0.1235 | 0.3911 | 0.0081 |

Table 12: Exploratory Critic flag rates (XES3G5M fold 0, $n$=500, seed 42, Qwen2.5-7B via Ollama unless noted). v1–v2 are historical graph-only packs; the QKC-T grounded/IG pair is the reported-checkpoint rerun (Table 13). This table is not a predictive attribution result.

| Run | Backend | Flag rate | Mean $P$(correct) |
|---|---|---|---|
| Stub (sanity) | StubLLM | 0.000 (0/500) | 0.742 |
| Ollama v1 | qwen2.5:7b | 0.124 (62/500) | 0.746 |
| Ollama v2 (format calib.) | qwen2.5:7b | 0.000 (0/500) | 0.753 |
| Grounded + `GroundedKCs` (v2) | qwen2.5:7b | 0.026 (13/500) | — |
| IG (v2, w/o $P$ in Diag.) | qwen2.5:7b | 0.768 (384/500) | — |
| Grounded + Critic (QKC-T) | qwen2.5:7b | 0.018 (9/500) | — |
| IG (QKC-T) | qwen2.5:7b | 0.766 (383/500) | — |

Table 13: Tier 2 explanation faithfulness: grounded Diagnostician vs. independent generation (IG). ID-anchored KC-Jaccard (JC) measures overlap between `GroundedKCs` trailer IDs and the support set $\{\text{target}\} \cup \text{history} \cup \text{1-hop } E_{\text{pre}}$. IG omits frozen $P(\text{correct})$ from the Diagnostician prompt; the Critic still sees the true Tier-1 score. The original Ollama pack used the v2 graph-only checkpoint (also the human-rating pack). The reported Q←KC+$\Delta t$ checkpoint is a same-prompt re-run ($n$=500).

| Arm | Backbone | Backend | $n$ | Flag | Mean JC |
|---|---|---|---|---|---|
| Grounded | — | StubLLM | 20 | 0.000 | 0.191 |
| IG | — | StubLLM | 20 | 1.000 | 0.191 |
| Grounded | v2 graph-only | qwen2.5:7b | 500 | 0.026 | 0.152 |
| IG | v2 graph-only | qwen2.5:7b | 500 | 0.768 | 0.142 |
| Grounded | v4 Q←KC+$\Delta t$ | qwen2.5:7b | 500 | 0.018 | 0.161 |
| IG | v4 Q←KC+$\Delta t$ | qwen2.5:7b | 500 | 0.766 | 0.152 |

Table 14: Human rating of exploratory Tier 2 outputs from the QKC-T checkpoint (XES3G5M fold 0, grounded Ollama subsample, $n$=100 explanations, 3 independent raters, Likert 1–5). Agreement: ordinal Krippendorff $\alpha$ and the mean of pairwise Spearman $\rho$ / fraction of scores within $\pm 1$.

| Scale | Mean ± SD | Agreement (3 raters) |
|---|---|---|
| Faithfulness | $4.55 \pm 0.63$ | $\alpha$=0.52; $\rho$=0.43; within±1: 95.0% |
| Usefulness | $4.01 \pm 0.74$ | $\alpha$=0.56; $\rho$=0.34; within±1: 94.7% |

Table 15: Paired ablation of DH$^2$A-KT knowledge and explanation knobs. Predictive rows use the clean $L$=400 protocol (fold 0, test-only) except HypergraphConv / FA session (validation or secondary corpus, as marked). The pre-specified gate is $\Delta$val $\geq$ +0.002; FAIL rows are not AUC novelty. Manipulation $\Delta$AUC is the v2 graph-only diagnostic (Appendix A), not the recommended tracer.

| Setting | AUC | Gate / note | Flag | Mean JC |
|---|---|---|---|---|
| Q←KC +$\Delta t$ (recommended) | 0.8296±0.0001 | 5/5 PASS +0.00274 | — | — |
| Q←KC on | 0.8270±0.0002 | not an incidence credit | — | — |
| Q←KC off (twin) | 0.8206±0.0002 | mismatched | — | — |
| HypergraphConv (val) | 0.8262 | FAIL +0.00025 | — | — |
| Grounded + Critic (v2 ckpt) | 0.752±0.002 | manip. 0.038–0.047 | 0.026 | 0.152 |
| IG (v2) | — | — | 0.768 | 0.142 |
| Grounded + Critic (v4 +$\Delta t$) | 0.8296 | same prompt, $n$=500 | 0.018 | 0.161 |
| IG (v4 +$\Delta t$) | — | — | 0.766 | 0.152 |
| w/o Critic | — | — | 0.000 | 0.152 |
| FA: concept-only | 0.720 | — | — | — |
| FA: + session HE | 0.722 | L FAIL +0.00007 | — | — |

Table 16: Ground-truth cross-validation on Junyi Academy (fold 0, top-$K$=\|expert\|=1131). Hyperedge chain slightly underlaps pairwise $E_{\text{pre}}$ but remains the same order of magnitude.

| Representation | Edge F1 | Edge prec. | Edge rec. |
|---|---|---|---|
| P0 $E_{\text{pre}}$ | 0.183 | 0.183 | 0.183 |
| Hyperedge chain | 0.165 | 0.165 | 0.165 |

Table 17: FoundationalASSIST secondary corpus (fold 0). Session hyperedges use a 30-minute gap; Hint members appear when `hint_count`$> 0$. Predictive AUC is observational (no P0-reused GKT/simpleKT column). An IPW identification check of hint use is in Appendix C; it is not a contribution.

| Quantity | Value |
|---|---|
| Interactions (canonical) | 1,740,987 |
| Learners / items / KCs | 5,000 / 3,359 / 209 |
| Hint-use rate | 0.052 |
| Train-only $E_{\text{pre}}$ edges (after cycle prune) | 280 |
| Path-derived concept-group chain hyperedges | 5,840 |
| Session hyperedges (train users) | 308,670 |
| with $\geq$1 Hint member | 35,752 (11.6%) |
| Session hyperedge ECR$^{\text{flag}}$ / group leak | 0 / 0 |
| DH$^2$-KT fold 0 AUC (path-derived concept-group) | 0.720 |
| + session HE (20k, Hint-prefer) | 0.722 ($\Delta$+0.002) |
| $n$ next-step predictions | 298,500 |

## 7. Discussion

### *7.1. Threats to validity*

Headline AUCs (Table 4) are clean $L$=400, test-only. Q←KC+$\Delta t$ is also on folds 1–2; compute-matched GIKT/AKT remain fold 0. Other baselines come from reused P0 tables on the older budget. The GIKT edge is small; we report learner-level bootstrap CIs, not a SOTA headline. HypergraphConv and hint-item hyperedges fail the pre-specified gate, so we do not credit multi-way message passing. The node-drop check (Appendix Appendix A) is run on the graph-only diagnostic encoder, not on the recommended tracer. Node-drop confirms *dependence* on some graph input. Degree-preserving rewiring on folds 1–2 additionally moves AUC (0.0341/0.0351), which is evidence that this diagnostic encoder is sensitive to co-occurrence structure, not only to the presence of nodes. That check still does not support graph-structure claims about QKC-T. Q←KC incidence is a bipartite membership matrix, not a new multi-way graph.

*Deployment availability of the query-side gap..* Attempt-onset scoring (this paper's protocol) produces $P(y_{t+1}{=}1)$ when the learner opens item $t$+1, so the query-side wait has already elapsed. History-only / recommend-before-return uses only LSTM-side gaps already in the log; that arm is the history-only row (five-seed mean $\Delta$val FAIL, 2/5 PASS; Table 10). Serving a next item before the learner returns is earlier still: the wait has not occurred, so query-side $\Delta t$ must be omitted or replaced by a planned-gap proxy we do not evaluate. The recommended tracer therefore assumes attempt-onset availability; a history-only deployment should use the LSTM-only arm and must not claim the both-mode 5/5 credit (mean $\Delta$val+0.00274).

### *7.2. Interpretation of the empirical findings*

The results answer an attribution question, not "hypergraph NNs beat GIKT." Each paragraph below asks whether one knowledge type survived the pre-specified ablation, and what that means for a system builder.

*Question–KC incidence..* Does not survive as credited structural knowledge under the capacity-matched twin (0/5 PASS; Table 5). The recommended tracer still includes the GIKT-class question pathway for architecture parity, but the train-only incidence message itself is not what the gate attributes.

*Linear* $\log(1+\Delta t)$.. Survives on this backbone when both LSTM-side and query-side gaps are present (5/5 PASS vs. Q←KC; 5/5 PASS vs. zero incidence; Table 6). History-only $\Delta t$ fails the gate (Table 10): the credited timing is attempt-onset context, not a history-only forgetting cue.

*Hypergraph convolution and path-derived concept groups..* Do not survive as predictive knowledge: `HypergraphConv` on chain hyperedges fails the gate (Table 9). Those groups remain an audit/diagnostic artefact. The ≈8 point gap versus GKT is the cost of the graph-only diagnostic encoder, not of QKC-T.

*Hint-item hyperedges, full Q-matrix, teacher grouping, expert DAG, exponential forgetting..* Do not survive (Table 9). Session+Hint on FoundationalASSIST (0.720 → 0.722) is a small local shift, not credited as multi-way graph novelty.

The Critic survey in Section 6.4 is exploratory and is not part of this attribution map.

*7.3. Limitations*

- Discussion-thread and teacher-intervention hyperedges remain data-blocked; session+Hint hyperedges on FoundationalASSIST shift AUC 0.720 $\rightarrow$ 0.722 under concept projection, but the pre-specified hint-item hypergraph arm fails the +0.002 gate (L in Table 9) and is not claimed as hypergraph novelty.

- Appendix Appendix C reports an IPW hint check with weak positivity; it is not a contribution and must not be read as a recommendation against hints.

- The hyperedge leakage audit's pairwise-projection approximation (Section 4.2) weights all projected pairs equally; a hyperedge-native statistic is future work.

- Exploratory Critic survey: one fold, one local 7B LLM. Blinded Likert on the recommended tracer ($n$=100, 3 raters) is reported in Table 14; ordinal Krippendorff $\alpha$ is 0.52 (faithfulness) / 0.56 (usefulness). Mean pairwise Spearman is moderate ($\rho\approx$0.43/0.34) while within$\pm$1 agreement is high ($\approx$95%), consistent with compressed high scores. An earlier dual-rater $n$=40 pack on the v2 graph-only checkpoint is archived only. The grounded–IG flag gap is largely definitional because IG withholds $P$(correct) from the Diagnostician while the Critic scores against that number. ID-anchored KC-Jaccard is not synonym-aware (XES3G5M lacks an in-repo KC name vocabulary); the JC gap is $\approx +0.010$. An LLM-scale sensitivity sweep remains optional future work.

- Reproduction environment: Windows workstation, single RTX 3090; unit tests pass in the released codebase.

## 8. Conclusion and future work

We presented DH$^2$A-KT as leakage-audited knowledge attribution: which structural and temporal knowledge survives a controlled ablation. Linear $\Delta t$ survives on the QKC-T backbone and on a capacity-matched zero-incidence twin; capacity-matched zeroing of the question–KC incidence message fails, as do hypergraph convolution and several metadata sources. On XES3G5M (clean $L$=400, test-only), the recommended tracer reaches 0.8296±0.0001 over five training seeds on fold 0 versus compute-matched GIKT 0.8258±0.0001 (0.8295±0.0004 over three learner folds). History-only $\Delta t$ does not pass the gate; only LSTM+query timing does. Future work: a tuning search for GIKT/AKT at $L$=400, and hyperedge-native leakage statistics.

## CRediT authorship contribution statement

**Tuan Dao Minh:** Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Khanh-Trinh Nguyen:** Validation, Data curation, Writing – review & editing. **Duong Nguyen Tien:** Validation, Data curation, Writing – review & editing. **Quoc Khanh Ngo:** Investigation, Resources, Writing – review & editing. **Van-Hau Nguyen:** Validation, Resources. **Le Hoang Son:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Ethics

The Likert study rates de-identified, model-generated explanations. It does not recruit learners, collect new educational records, or intervene in instruction. The three raters scored system output already produced from the public XES3G5M release. The authors' institutions do not require research-ethics board review for this form of expert rating of model output.

## Data availability

Code matching this manuscript is at `https://github.com/edu-risk-lab/DH2A-KT` (default branch, commit `b60e6b8`; a Zenodo snapshot DOI will be minted at submission). The repository will be made public at submission; until then, reviewers may request access from the corresponding author. P0's audited edges and reused baselines are vendored under [8]'s release terms (pinned commit `5e3d6a0`, Section 5.2); the companion PDF can be supplied as journal supplementary material while that manuscript is under review. FoundationalASSIST [9] must be obtained from Hugging Face under its Responsible Use Guidelines and cannot be redistributed by the authors. A table-to-script map is in the repository file `docs/paper-artifacts.md`.

**Acknowledgment**

All Tier 1 experiments in this paper are designed to run on a single consumer-grade NVIDIA RTX 3090 (24GB), matching the compute budget already validated by [8] on the same primary benchmarks. FoundationalASSIST [9] was accessed under the Hugging Face dataset's Responsible Use Guidelines for this study.

## Appendix A. Graph-only encoder as a forced-reliance diagnostic

The recommended tracer is QKC-T (Table 4). A graph-only diagnostic encoder (chain hyperedges, no question LSTM path; log label v2) exists so a destruction check is interpretable: mean test AUC 0.7516±0.0015 vs. reused GKT 0.8336±0.0010 under P0's GKT budget ($L$=200; Table A.18). Frozen-weight node-drop $p$=0.9 passes on all three folds ($\Delta$AUC 0.038/0.043/0.047); P0 GKT also drops after retraining on a destroyed pairwise graph (0.0877±0.0009). Protocol differs (freeze vs. retrain). Node-drop confirms dependence on some graph input. Degree-preserving rewiring passes on folds 0–2 ($\Delta$AUC 0.0318/0.0341/0.0351; clean AUC $\approx$ 0.75). Relation-label permutation is near-null on every fold (Table A.19). We do not deploy this encoder.

Table A.18: Diagnostic graph-only track (not QKC-T or Zero+$\Delta t$). Test AUC on XES3G5M under P0's matched GKT budget (batch 4, 10 epochs, max seq. len. 200). DH$^2$-KT v2 uses path-derived concept-group chain hyperedges and graph-only interaction. Mean±SD over three learner folds.

| Fold | DH$^2$-KT v2 | GKT (P0) | simpleKT (P0) |
|---|---|---|---|
| 0 | 0.7501 | 0.8346 | 0.8744 |
| 1 | 0.7518 | 0.8338 | 0.8747 |
| 2 | 0.7531 | 0.8326 | 0.8750 |
| **Mean±SD** | **0.7516±0.0015** | **0.8336±0.0010** | **0.8747±0.0003** |
| $\Delta$ vs. GKT (mean±SD) | | −0.0820±0.0025 | |

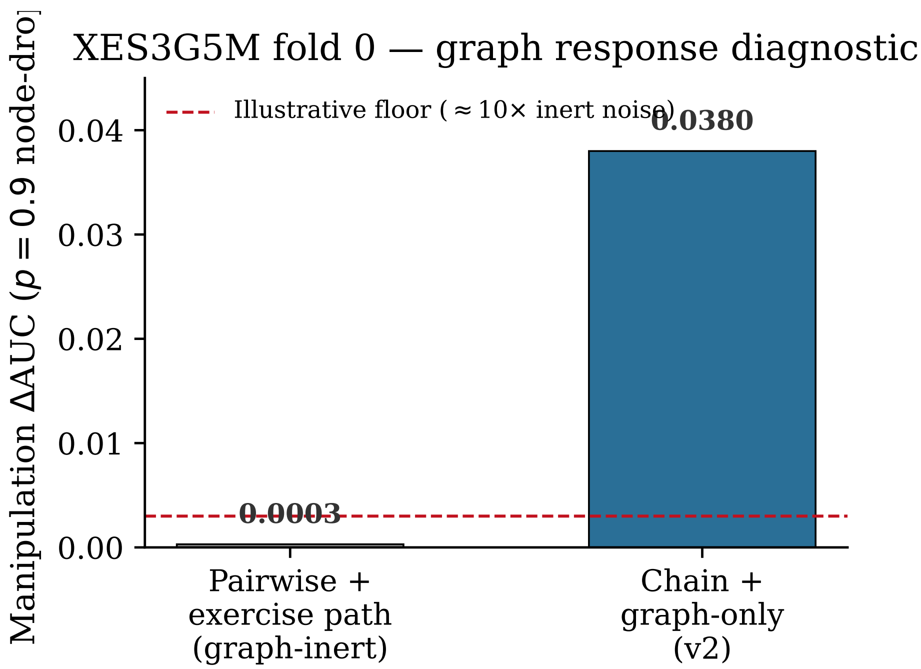


Figure A.3: Manipulation-check $\Delta$AUC on XES3G5M fold 0 under $p$=0.9 node-drop. An early pairwise+exercise-embedding backbone is graph-inert ($\Delta$AUC≈ 0.0003); the chain+graph-only redesign moves ($\Delta$AUC= 0.038). The dashed line at 0.003 is an *illustrative* floor (≈10× the graph-inert noise), not a formal statistical cutoff from [8].

Table A.19: Manipulation check on XES3G5M. Reported destruction is node-drop $p$=0.9 on the graph-only diagnostic encoder (log label v2): train on clean hyperedges, re-evaluate frozen weights on destroyed hyperedges (folds 0–2). This confirms dependence on some graph input, not correct use of relation structure. Degree-preserving rewiring and relation-label permutation (Hau B12) use the same frozen checkpoints (`xes3g5m_fold{0,1,2}.pt`). Rewiring passes on all three folds ($\Delta$AUC 0.0318–0.0351; clean AUC $\approx$ 0.75, matching the node-drop encoder); label permutation is near-null (expected for a single-kind encoder). GKT: P0 DDR-downstream mean $\Delta$AUC over $n$=9 fold–seed runs ([8]; retrain on destroyed pairwise graph). simpleKT has no concept-graph input, so graph destruction is undefined. QKC-T and Zero+$\Delta t$ do not take this check.

| Model / destruction | AUC clean | $\Delta$AUC | DDR | Pass? |
|---|---|---|---|---|
| Diagnostic encoder, node-drop $p$=0.9 (fold 0) | 0.753 | 0.0380 | 0.990 | Yes |
| Diagnostic encoder, node-drop $p$=0.9 (fold 1) | 0.749 | 0.0432 | 0.990 | Yes |
| Diagnostic encoder, node-drop $p$=0.9 (fold 2) | 0.753 | 0.0469 | 0.990 | Yes |
| Diagnostic encoder, degree-preserving rewiring (fold 0) | 0.749 | 0.0318 | 0.057 | Yes |
| Diagnostic encoder, degree-preserving rewiring (fold 1) | 0.751 | 0.0341 | 0.058 | Yes |
| Diagnostic encoder, degree-preserving rewiring (fold 2) | 0.752 | 0.0351 | 0.057 | Yes |
| Diagnostic encoder, relation-label permutation (fold 0) | 0.749 | $\approx$0 | 0.000 | No |
| Diagnostic encoder, relation-label permutation (fold 1) | 0.751 | $\approx$0 | 0.000 | No |
| Diagnostic encoder, relation-label permutation (fold 2) | 0.752 | $\approx$0 | 0.000 | No |
| GKT (P0, mean$\pm$SD), node-drop $p$=0.9 | — | 0.0877$\pm$0.0009 | 0.990 | Yes |
| simpleKT (P0) | — | N/A | — | N/A |

## Appendix B. Hyperparameters for Table 4

Table B.20 records, for each model in Table 4, which knobs were searched, which value was used, and how many distinct configurations were tried [29]. We do not claim a matched *tuning* budget.

Table B.20: Training recipes for every model in Table 4. "Configs tried" counts distinct hyperparameter vectors, not seeds. DH$^2$-KT's *architecture* was chosen from a pre-specified ablation ladder (Table 9 plus Q$\leftarrow$KC and three $\Delta t$ injection sites); its *optimiser* recipe is a single vector. GIKT/AKT/*simpleKT*/DKT/DGEKT each used one inherited PYKT vector [26]. AKT and *simpleKT* were not retuned when $L$ moved from P0's 200 to 400. GIKT seed 42 (log `C_gikt_clean_L400_e30b16`) restored best valid AUC at epoch 8 and stopped by epoch 13 under patience 5 (cap 30).

| Model | Parameter | Range searched | Selected | How chosen ($n$ configs) |
|---|---|---|---|---|
| DH$^2$-KT tracer | architecture | ablation ladder | v4 Q$\leftarrow$KC +$\Delta t$ both | pre-specified gate |
| DH$^2$-KT tracer | batch / epochs / lr / $L$ / patience | — | 16 / 30 / $10^{-3}$ / 400 / 5 | one recipe × 5 seeds |
| GIKT | batch / epochs / patience / $L$ | — | 16 / 30 / 5 / 400 | match DH$^2$ compute ($n$=1) |
| GIKT | remaining pyKT hparams | none | P0/pyKT defaults | inherited ($n$=1) |
| AKT | $L$ | {400} override | 400 | L matched; other hparams inherited ($n$=1) |
| AKT | lr / dropout / d_model | none at $L$=400 | P0/pyKT defaults | inherited from $L$=200 recipe |
| *simpleKT* | $L$ / other hparams | {400} override | 400 + P0 defaults | inherited ($n$=1, 5 seeds) |
| DKT, DGEKT | $L$ / other hparams | {400} override | 400 + P0 defaults | inherited ($n$=1, 5 seeds) |

## Appendix C. Observational hint identification check

On FoundationalASSIST we report a standard inverse-propensity (IPW) average treatment effect of $\texttt{hint\_used}_t$ on $\texttt{correct}_{t+1}$, with logistic propensity on pre-$t$ summaries and clipping. Unconfoundedness is assumed, not proven. Same-step `discrete_score` is not the outcome (it is mechanically tied to hint requests). No estimate is reported on XES3G5M (no hint logs). This check is not a contribution.

Using train transitions, the IPW estimate of hint use on *next-step* correctness is −0.207 (naive mean difference −0.255; trimmed IPW −0.211; bootstrap 95% CI [−0.230, −0.187] on 50k-row resamples; $n_{\text{treated}}$=36,151). Propensity scores hit the clip floor ([0.05, 0.219]), indicating weak overlap. The negative sign reflects that weaker learners request more hints; it must *not* be read as the effect of providing hints, and it is not a recommendation against hints. We did not compute Rosenbaum bounds or an E-value.

Table C.21: FoundationalASSIST fold 0 IPW identification check of $\texttt{hint\_used}_t$ on $\texttt{correct}_{t+1}$ under pre-$t$ confounders. Propensity mass hits the clip floor. The negative IPW estimate reflects that weaker learners request more hints; it is *not* the effect of providing hints.

| Quantity | Value |
|---|---:|
| Hint ATE (IPW) / naive mean diff | −0.207 / −0.255 |
| trimmed IPW (propensity 10–90%) | −0.211 |
| bootstrap 95% CI (subsample) | [−0.230, −0.187] |
| $n$ treated / control (train transitions) | 36,151 / 660,321 |
| propensity [min, max] | [0.05, 0.219] |

### Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Cursor (Composer) in order to draft and revise LaTeX wording, tables, and cover-letter text under author direction. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The Tier 2 Diagnostician/Critic/Tutor agents (Qwen2.5-7B via Ollama) are experimental components of the reported system, not manuscript-writing tools, and are described in Section 6.4.

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